Data Structures and Algorithms

LECTURE 08: BINARY TREES, BINARY SEARCH TREES

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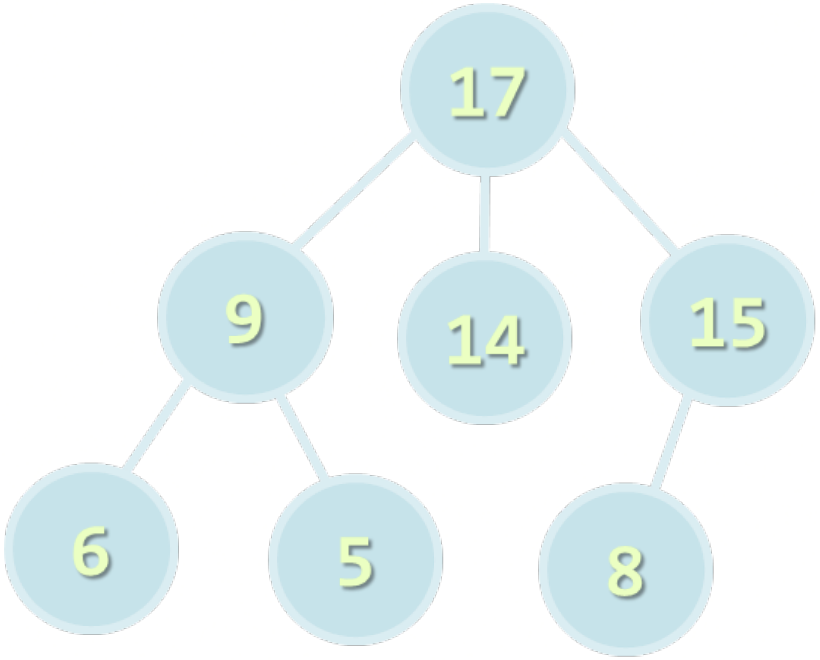
Alliance with FPT Education

Contents

- Binary Trees
  - Traversal algorithms
- Binary Search Trees

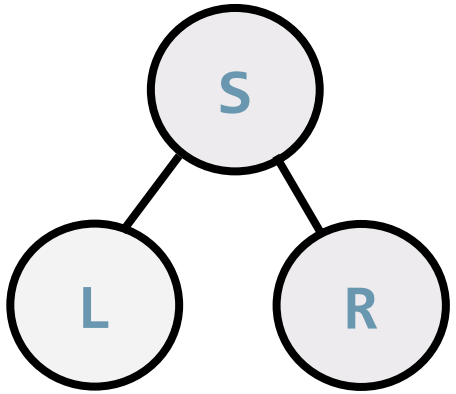
Binary Trees and BT Traversal



Preorder, In-Order, Post-Order

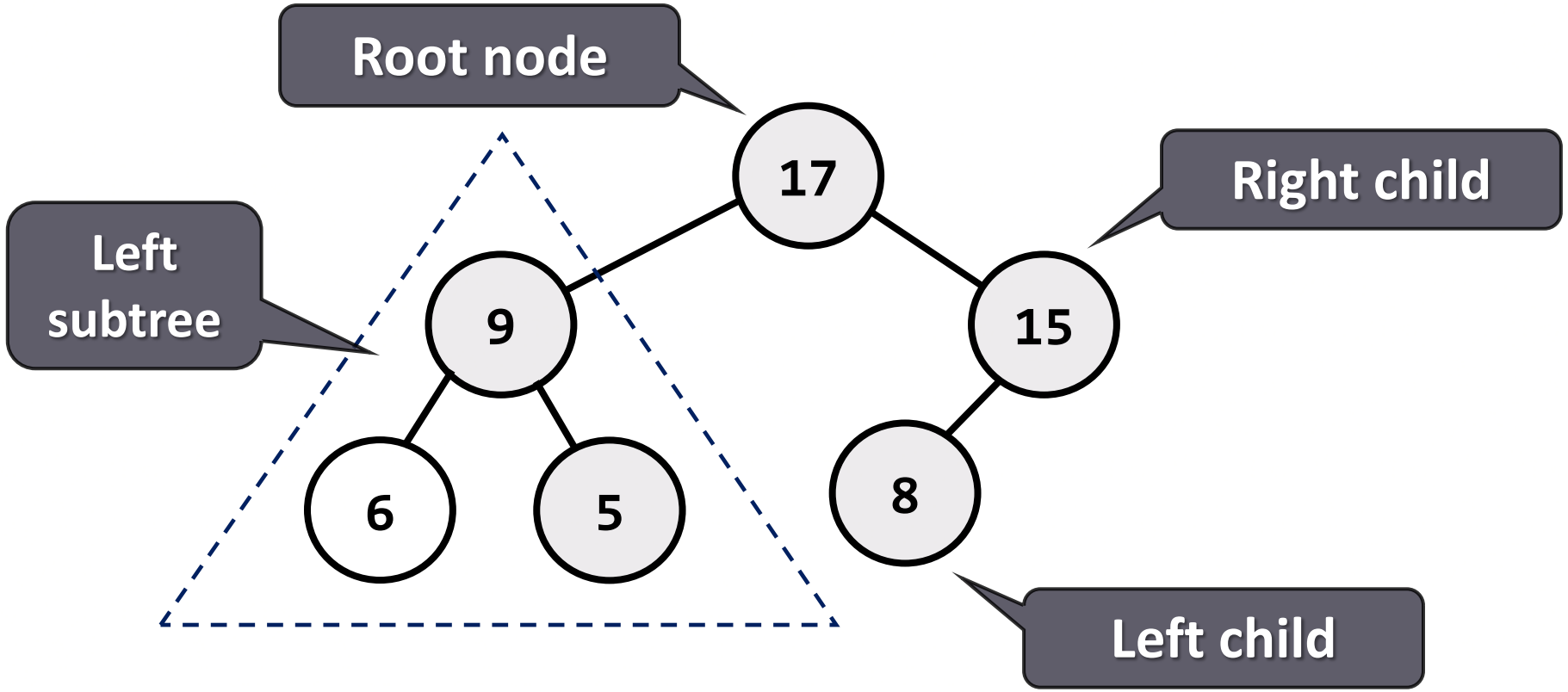
Binary Tree

- ADS representing tree like hierarchy
- Each node has **at most two** children
  - Children are called **left** and **right**
  - The **parent** is also called **source**



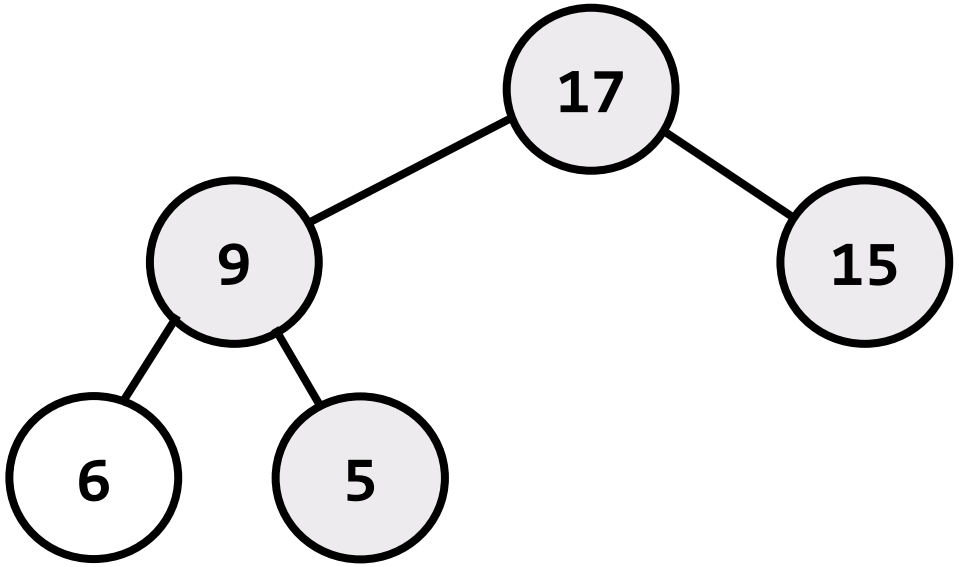
Binary Trees

- **Binary trees**: the most widespread form
  - Each node **has at most 2 children** (**left** and **right**)



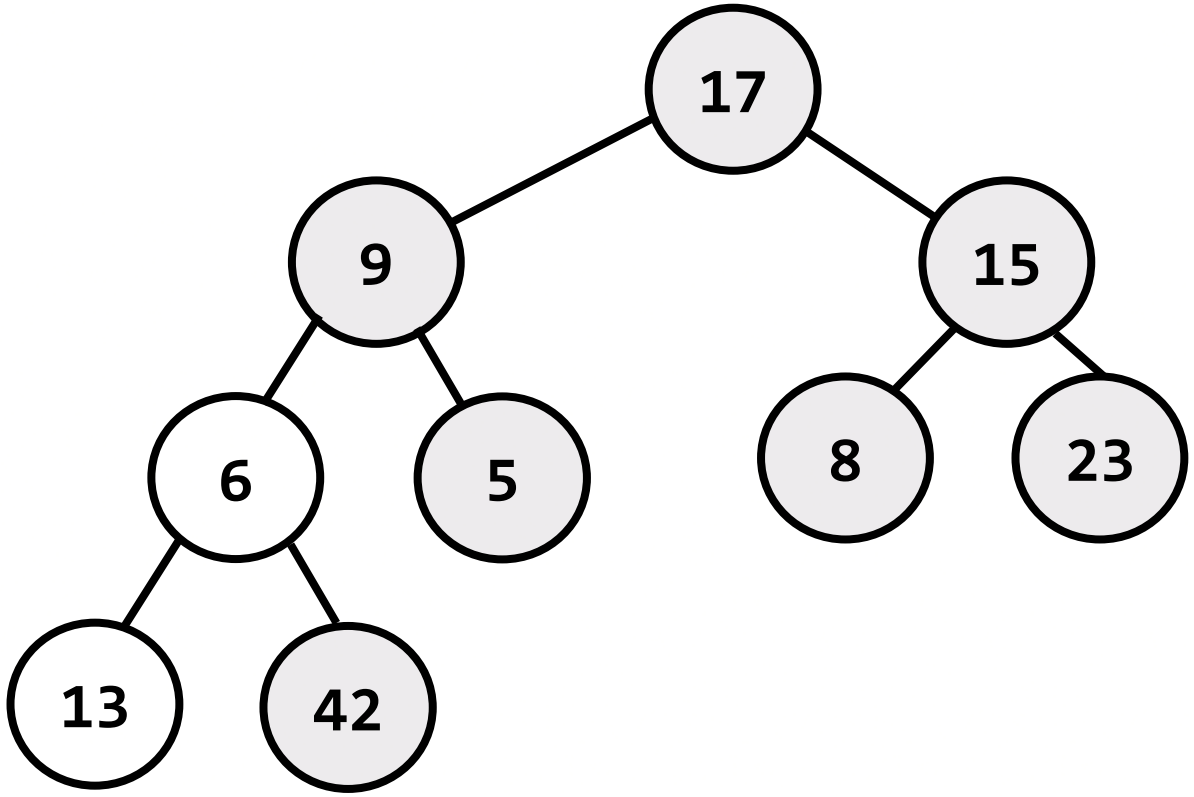
Types of Binary Trees

- **Full** – each node has **0** or **2** children



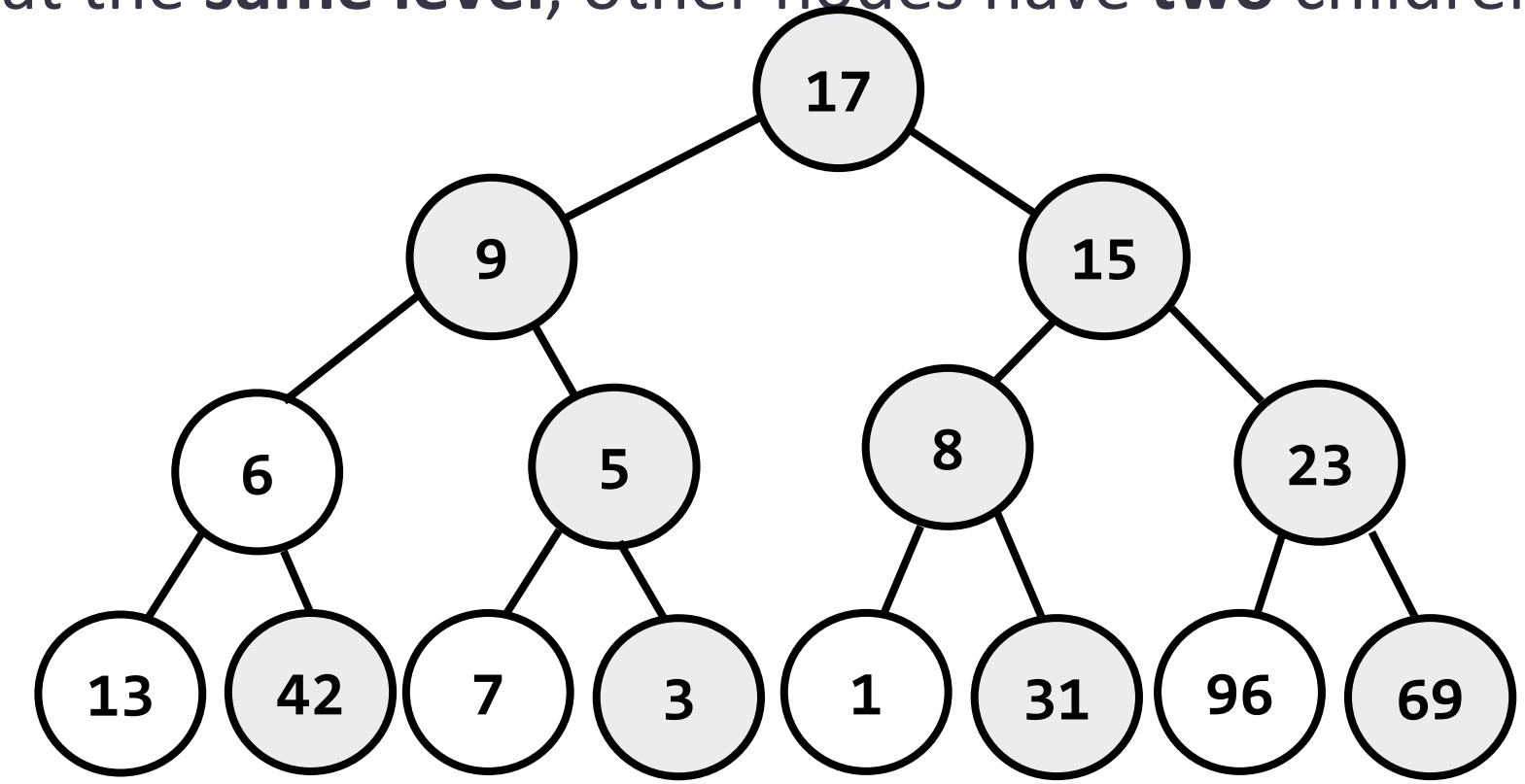
Types of Binary Trees

- **Complete** – nodes are filled **top** to **bottom** and **left** to **right**



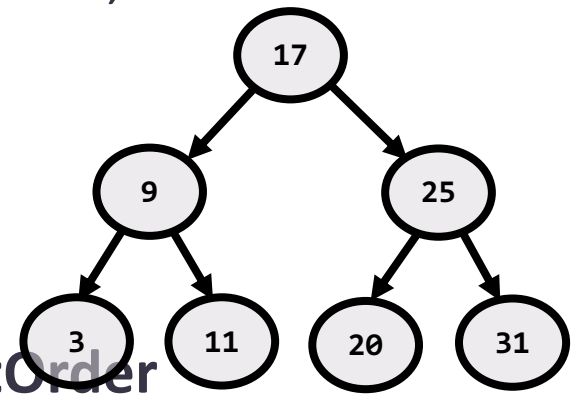
Types of Binary Trees

- **Perfect** – combines **complete** and **full**
  - leafs are at the **same level**, other nodes have **two** children



Problem: Binary Tree Traversals

- Inside the given skeleton
  - Implement **AbstractBinaryTree<E>**
  - Implement **asIndentedPreOrder**, level indented +2




```
17
  9
    3
    11
  25
    20
    31
```

each

- **preOrder**, **inOrder** and **postOrder**
  - Return the nodes as list List<**AbstractBinaryTree<E>**>

Solution: BT Traversals - Constructor

- Fields and constructor:

```
public class BinaryTree<E> implements AbstractBinaryTree<E> {
    private E key;
    private BinaryTree<E> left;
    private BinaryTree<E> right;

public BinaryTree(E key, BinaryTree<E> left, BinaryTree<E> right) {
    this.key = key;
    this.left = left;
    this.right = right;
}
```

Solution: BT Traversals - Print

```
public String asIndentedPreOrder(int indent) {
    String out = createPadding(indent) + getKey();
    if (getLeft() != null) {
        out +="\n" + getLeft().asIndentedPreOrder(indent + 2);
    }
    if (getRight() != null) {
        out +="\n" + getRight().asIndentedPreOrder(indent + 2);
    }
    return out;
}
```

Process Node

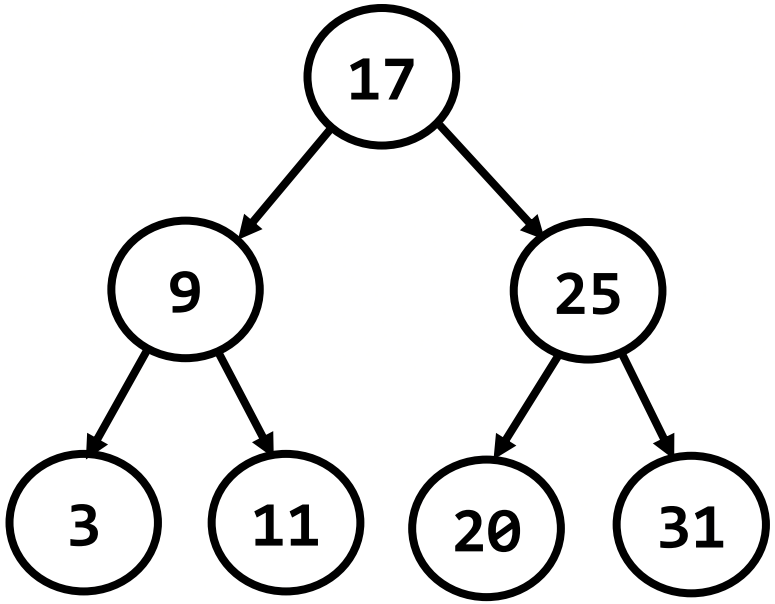
Traverse Left

Traverse Right

Binary Trees Traversal: Pre-order

- Root → Left → Right

```
preOrder (node) {
  if (node != null) {
    print node.key
    preOrder(node.left)
    preOrder(node.right)
  }
}
```

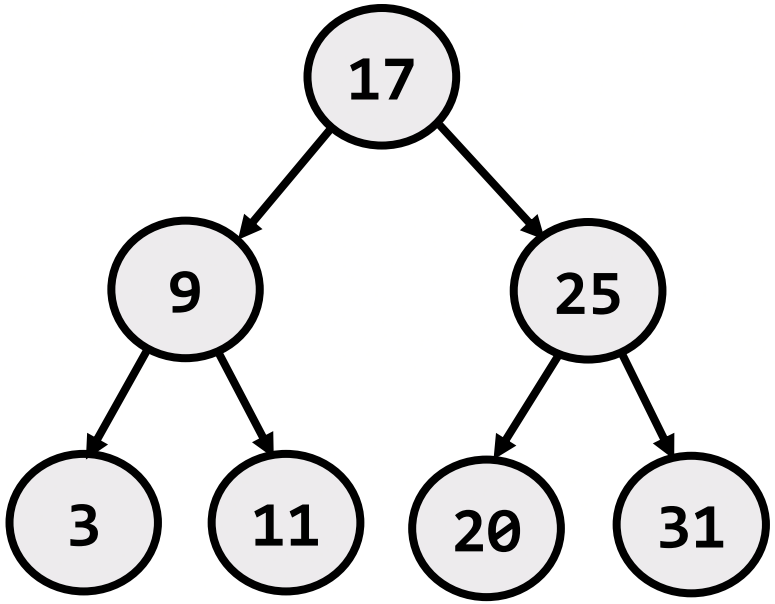


→ 17 9 3 11 25 20 31

Binary Trees Traversal: In-order

- Left → Root → Right

```
inOrder (node) {
  if (node != null) {
    inOrder(node.left)
    print node.key
    inOrder(node.right)
  }
}
```

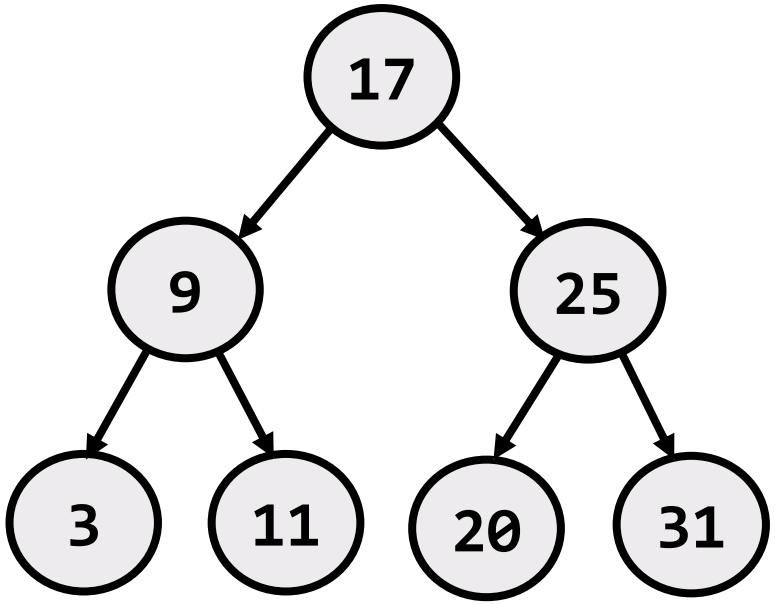


→ 3 9 11 17 20 25 31

Binary Trees Traversal: Post-order

- Left → Right → Root

```
postOrder (node) {
  if (node != null) }
    postOrder(node.left)
    postOrder(node.right)
    print node.key
  }
}
```

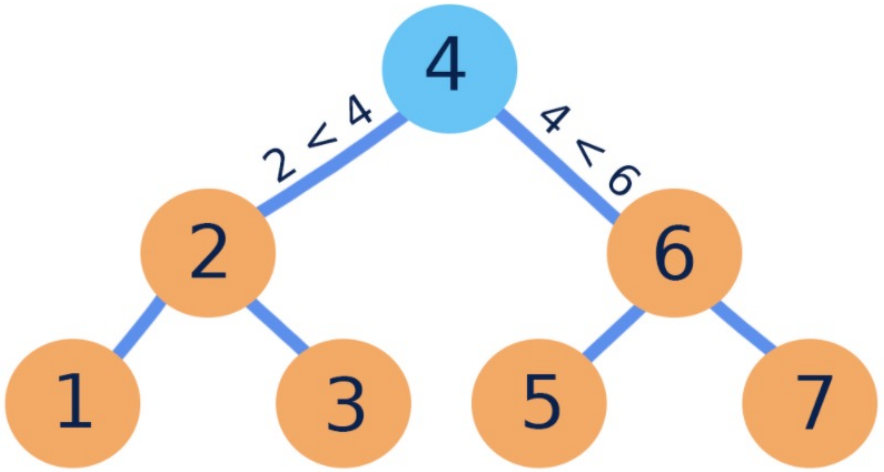


→ 3 11 9 20 31 25 17

Solution: BT Traversals - forEachInOrder

```
public void forEachInOrder(Consumer<E> consumer) {
    if (this.getLeft() != null) {
        this.getLeft().forEachInOrder(consumer);
    }
    consumer.accept(this.getKey());
    if (this.getRight() != null) {
        this.getRight().forEachInOrder(consumer);
    }
}
```

Binary Search Trees



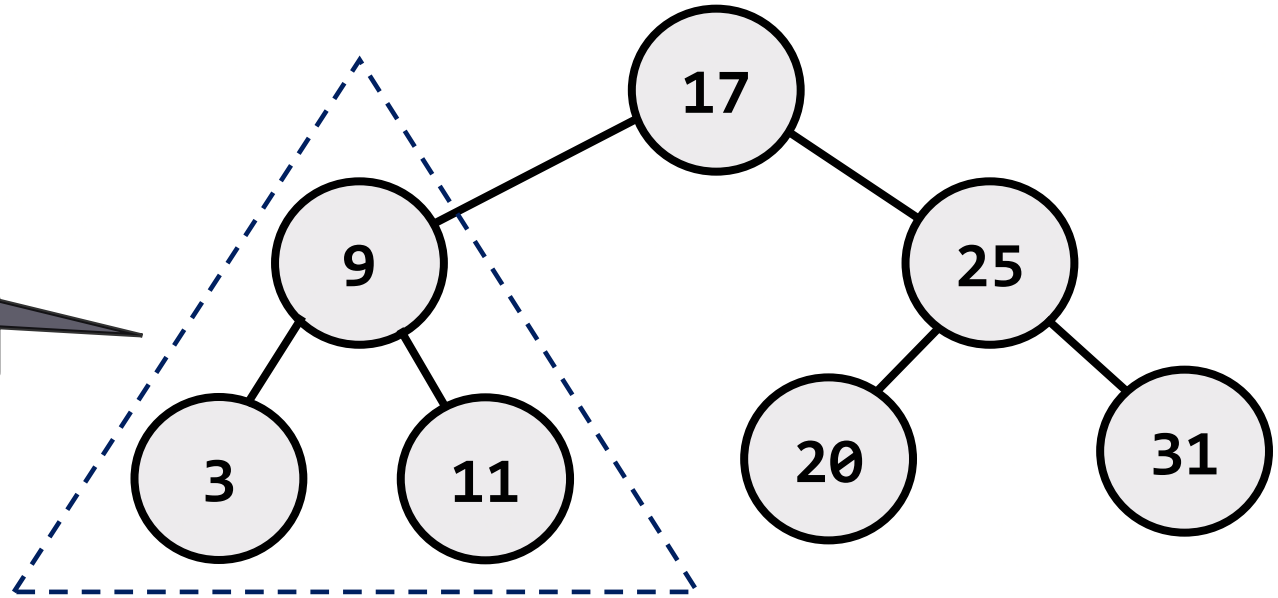
Two Children at Most

Binary Search Trees

- **Binary search trees** are **ordered**
  - For each node ***x***
    - Elements in left subtree of ***x*** are **< *x***
    - Elements in right subtree of ***x*** are **> *x***

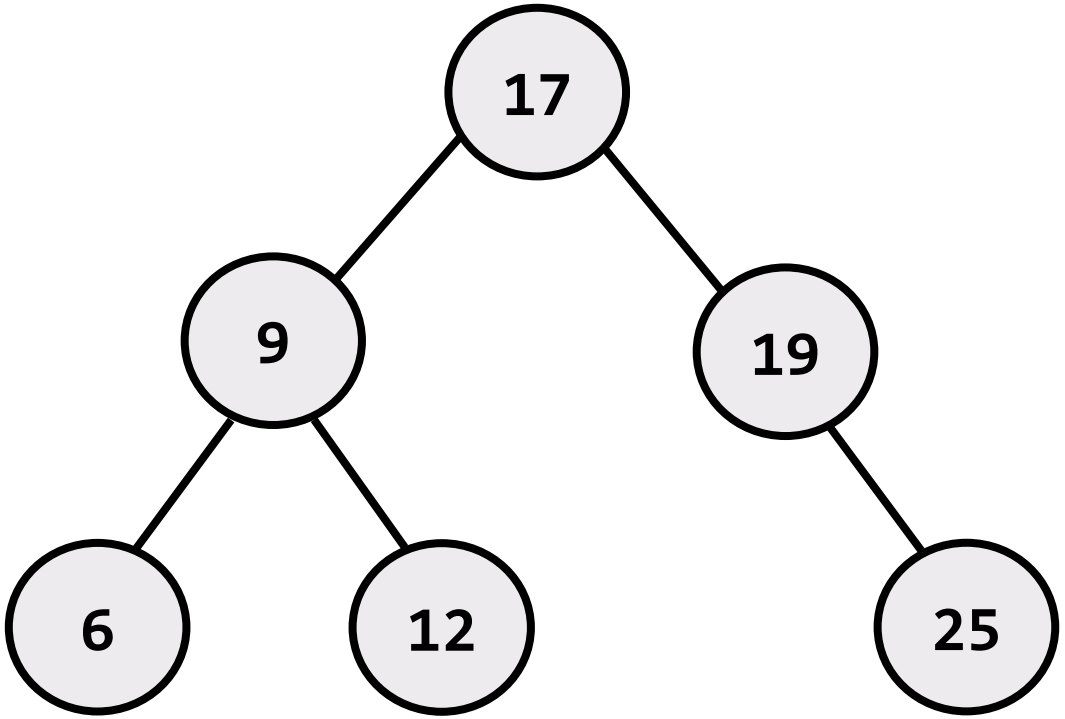
what about ==

nodes are < 17



BST - Search

- Search for **x** in BST
  - if node is not null
    - if x < node.value → **go left**
    - else if x > node.value → **go right**
    - else if x == node.value → **return**

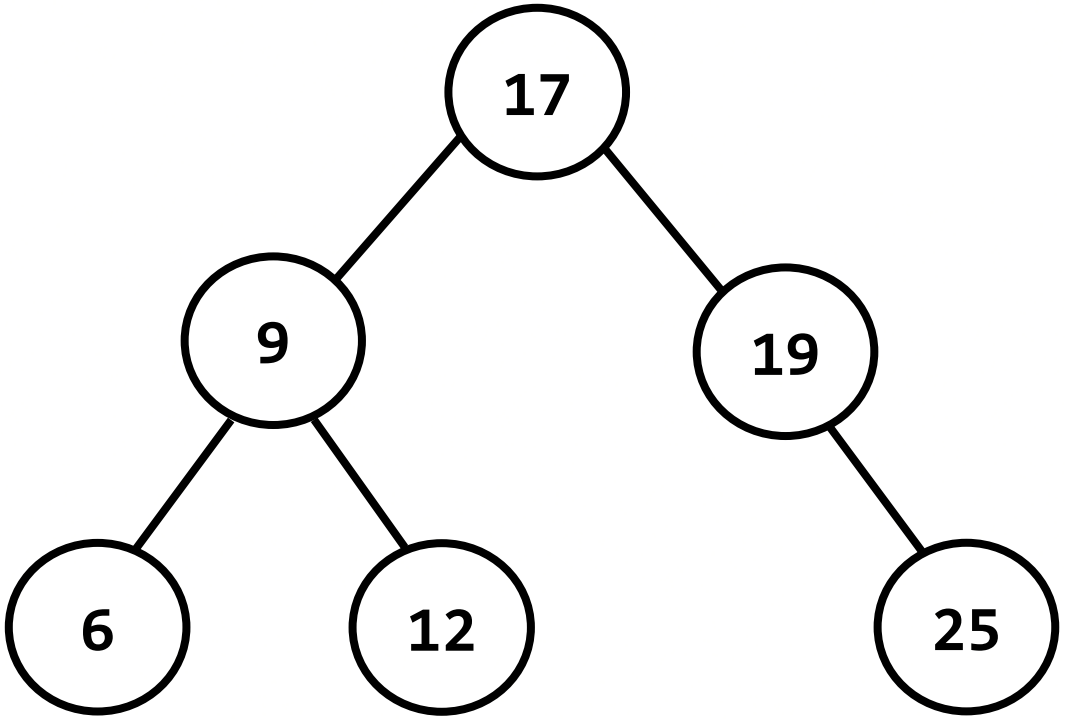


Search 12 → 17 9 12

Search 27 → 17 19 25 null

BST - Insert

- Insert **x** in BST
  - if node is **null** → insert x
  - else if x < node.value → **go left**
  - else if x > node.value → **go right**
  - else → node **exists**

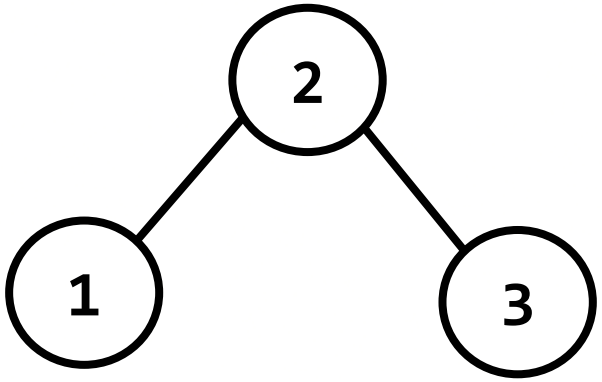


Insert 12 → 17 9 12 return

Insert 27 → 17 19 25 null(insert)

Problem: BST

- You are given a skeleton
  - Implement **AbstractBinarySearchTree<E>**
    - **bool contains(E element)**
    - **void insert(E element)**



Solution: BST Contains

```
public boolean contains(E element) {
    Node<E> current = this.root;
    while (current != null){
        if (element.compareTo(current.value) < 0){
            current = current.leftChild;
        } else if (element.compareTo(current.value) > 0){
            current = current.rightChild;
        } else {
            break;
        }
    }
    return current != null;
}
```

Solution: BST Insert

```
public void insert(E element) {
    if (this.root == null) {
        this.root = new Node<>(element);
    } else {
        // TODO: Find the place to insert
        if (parent.value.compareTo(element) > 0){
            parent.leftChild = new Node<>(element);
        } else {
            parent.rightChild = new Node<>(element);
        }
    }
}
```

Problem: BST Search

- Implement:
  - **BST<E> search(E value)**
- Make sure the method works for:
  - **empty tree**
  - tree with **one element**
  - tree with **two elements - root + left/right**
  - tree with **multiple elements**

Solution: BST Search

```
public AbstractBinarySearchTree<E> search(E element) {
   Node<E> current = this.root;
 // TODO: Find the node with the element
   return new BinarySearchTree<>(current);
}
```

Solution: BST Search (2)

```
private BinarySearchTree(Node<E> root) {
  this.copy(root);
}

private void copy(Node<E> node) {
  if (node == null) return;

  this.insert(node.value);
  this.copy(node.leftChildre);
  this.copy(node.rightChildren);
}
```

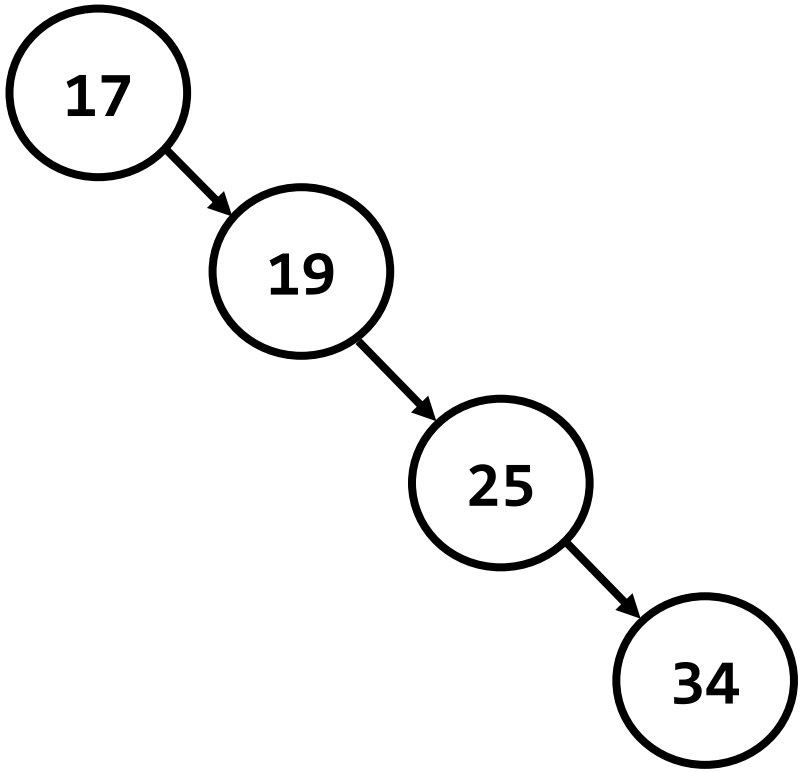
Pre-Order Traversal

BST - Search Operation Speed - Quiz

- What is the speed of the **search(E)** operation on BST?
  - O(n)
  - O(log(n))
  - O(1)

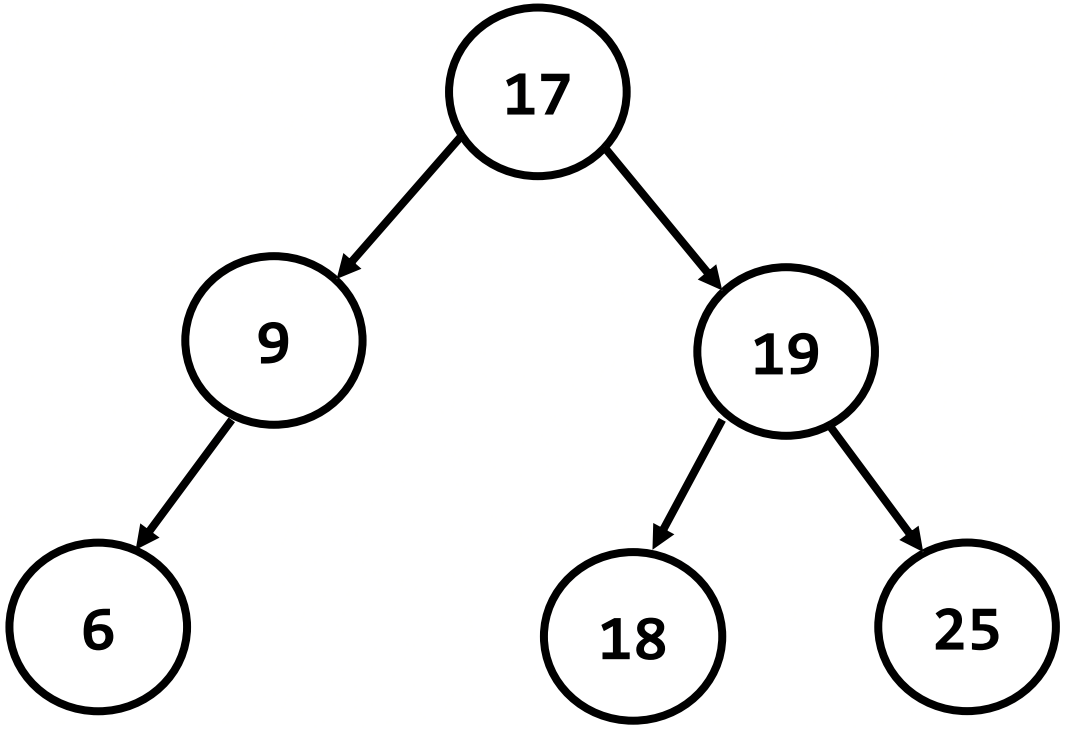
BST - Search Operation Speed - Answer

- What is the speed of the **search(E)** operation on BST?
  - O(n)
  - O(log(n))
  - O(1)



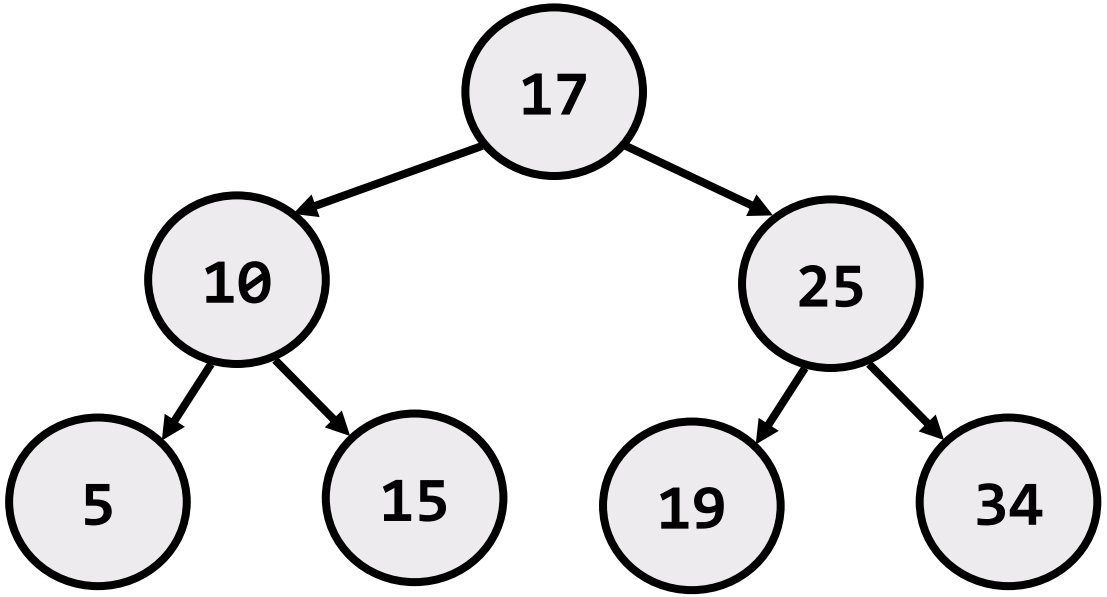
Binary Search Trees – Operation Speed

- Insert – **height** of tree
- Search – **height** of tree



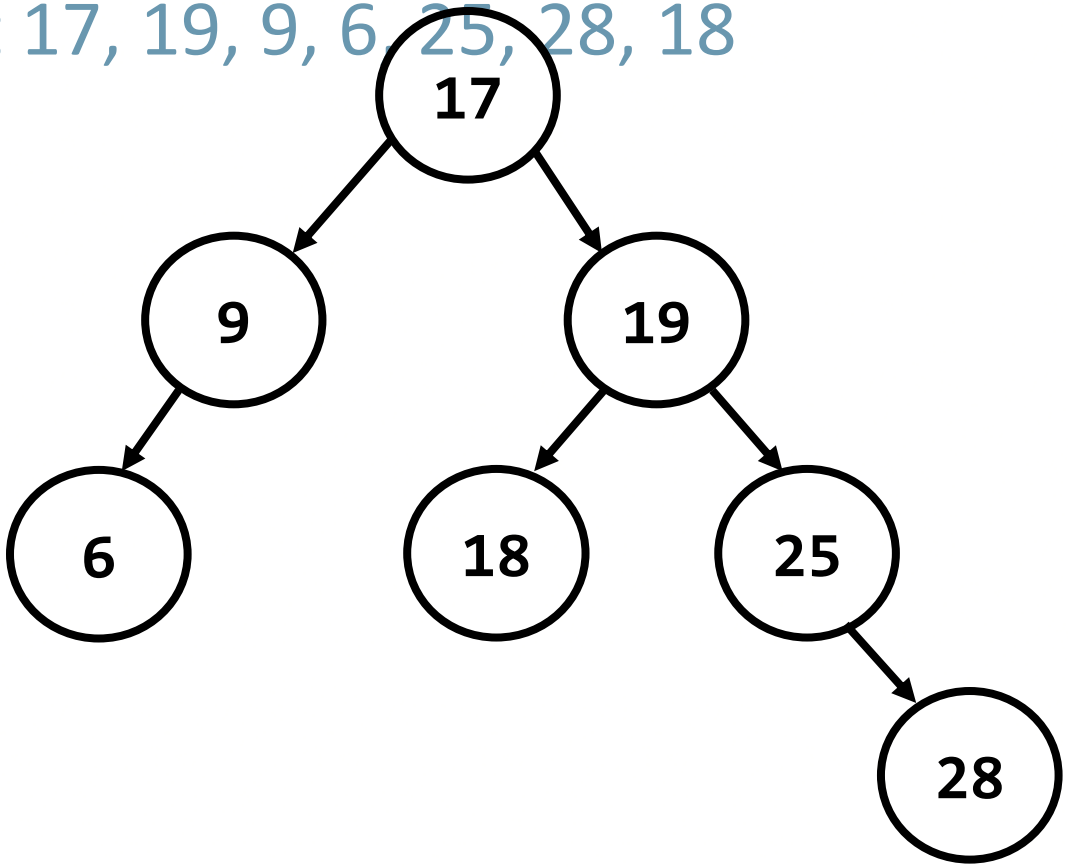
Binary Search Trees – Best Case

- Example: Insert 17, 10, 25, 5, 15, 19, 34



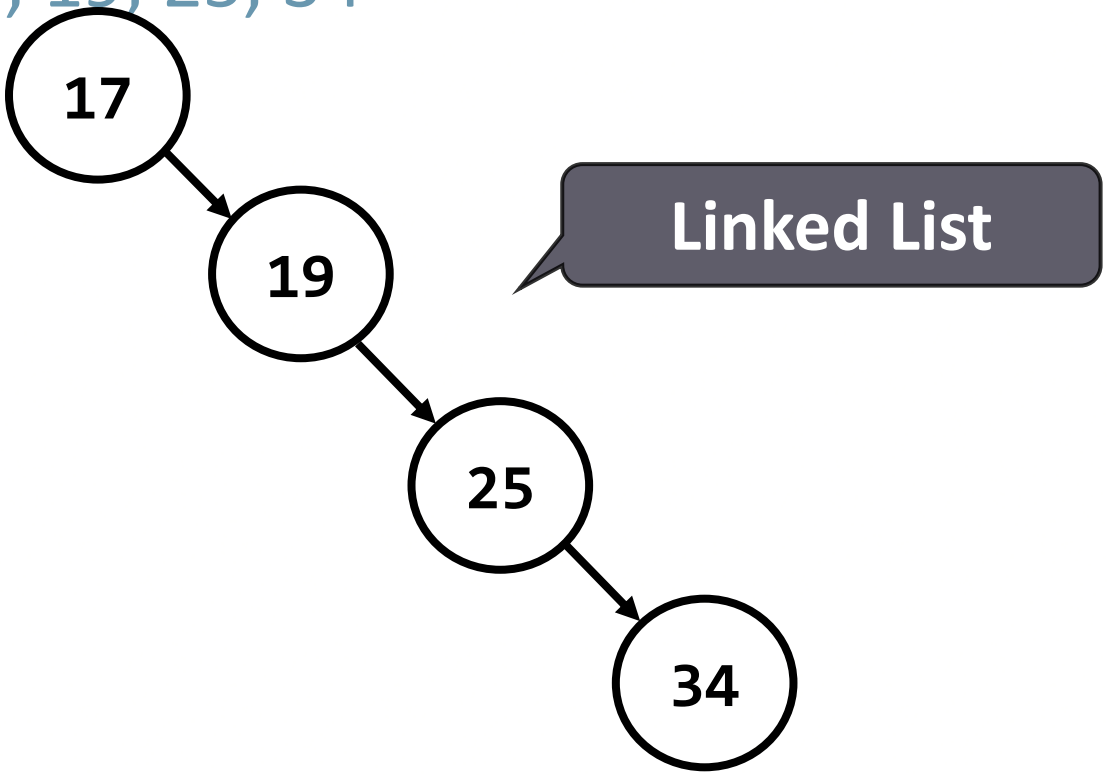
Binary Search Trees – Average Case

- You can insert values in ever **random** order
- Example: Insert 17, 19, 9, 6, 25, 28, 18



Binary Search Trees – Worst Case

- You can insert values in ever **increasing/decreasing** order
- Example: Insert 17, 19, 25, 34



Balanced Binary Search Trees

- Binary search trees can be **balanced**
  - Balanced trees have for each node
    - Nearly equal number of nodes in its subtrees
  - **Balanced trees** have **height of ~ log(n)**

Summary

- **Binary** trees have **0** or **2** children
- **Efficient** operations
  - **Add**
  - **Find** min
  - **Remove** min